

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:		Reg. No.:	
Date:		Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
3. **Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Problem Statement Domain: Employee Skill-Gap Analysis and Training Recommendation System — an AI-assisted HR platform that evaluates employee competencies against role requirements, identifies development needs from performance reviews, certifications, assessments and project experience, and recommends personalized training programs so that managers and HR teams can track workforce readiness through data-driven dashboards.

1 · Project Vision & Stakeholders

1.1 Project Vision

To build an intelligent, data-driven platform that continuously measures every employee's competencies against their role's skill requirements, closes identified gaps through personalized, ranked training recommendations, and gives managers and HR teams real-time visibility into workforce readiness through interactive dashboards — turning routine performance data into a future-ready, self-improving organization capable of adapting to fast-changing industry skill demands.

1.2 Project Objectives

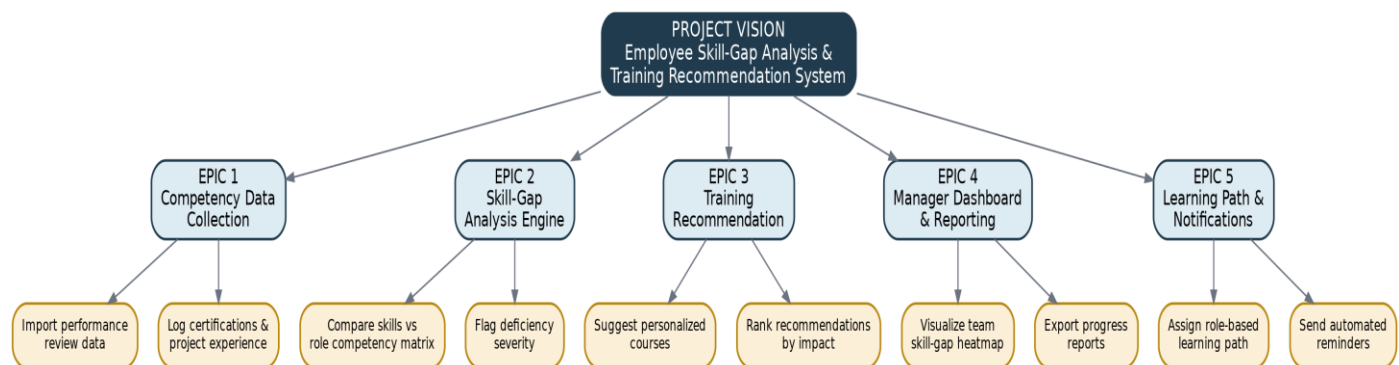
- Evaluate employee competencies using performance reviews, certifications, assessments, and project experience.
- Identify skill deficiencies against predefined job-role competency frameworks.
- Recommend personalized, role-aware training programs to close identified gaps.
- Track learning progress, certification status, and completion of assigned activities.
- Give HR and managers dashboards and analytical reports for workforce planning.
- Automate reminders/notifications to drive learning-activity completion.
- Enable scalable, data-driven decision-making in employee training and career development.

1.3 Key Stakeholders

Stakeholder	Category	Interest / Role in the System
Employee	Primary User	Views own skill profile, receives training recommendations, completes learning activities.
Reporting Manager	Primary User	Monitors team skill gaps, approves learning paths, reviews progress dashboards.
HR / L&D Team	Primary User	Defines competency frameworks, curates training catalogue, tracks org-wide metrics.
System Administrator	Secondary User	Manages roles, integrations with HRMS/LMS, and platform configuration.
Training Content Providers	External Partner	Supplies course catalogue and certification data via integration.
Senior Leadership	Sponsor	Consumes workforce-readiness analytics for strategic planning and budget decisions.
Recommendation Engine (AI module)	System Actor	Computes gap analysis and generates ranked training suggestions.

2 · Epics & User Stories

The product backlog is organized into five epics, each decomposed into user stories written in the standard Agile form: "As a <user>, I want <feature>, so that <benefit>."



Epic 1 — Competency Data Collection

This epic covers every entry point through which competency evidence enters the platform — self-reported certifications, imported performance-review scores, and manager endorsements. Reliable, current data at this layer is the foundation the gap-analysis engine depends on, so these stories are treated as prerequisites for Epic 2.

US-1.1 As an employee, I want to submit my certifications and project experience, so that my competency profile stays current.

US-1.2 As an HR admin, I want to import performance-review scores automatically, so that I don't have to re-enter existing evaluation data.

US-1.3 As a manager, I want to endorse an employee's demonstrated skill, so that peer/manager validation strengthens the profile's accuracy.

Epic 2 — Skill-Gap Analysis Engine

The analysis engine is the analytical core of the platform. It consumes the competency data collected in Epic 1 and a configurable competency framework per job role, then produces a structured gap report that every downstream feature — recommendations, dashboards, learning paths — relies upon.

US-2.1 As the system, I want to compare an employee's competency profile against the role's required competency matrix, so that gaps are identified automatically.

US-2.2 As an HR admin, I want to define and edit competency frameworks per department/role, so that gap analysis stays aligned with evolving job requirements.

US-2.3 As an employee, I want to view my own skill-gap report, so that I understand exactly which competencies I need to develop.

Epic 3 — Training Recommendation Engine

Once a gap is identified, this epic turns that finding into an actionable next step. The recommendation engine ranks catalogue courses by their expected impact on the specific competency gap, while giving employees the flexibility to accept or defer a suggestion around their existing workload.

US-3.1 As the system, I want to recommend training programs ranked by relevance to an identified skill gap, so that employees see the highest-impact courses first.

US-3.2 As an employee, I want to accept or defer a recommended course, so that I can plan learning around my workload.

US-3.3 As an HR admin, I want to map each course in the catalogue to one or more competencies, so that recommendations stay accurate as the catalogue grows.

Epic 4 — Manager Dashboard & Reporting

This epic surfaces the aggregated results of Epics 2 and 3 to the people who plan workforce development — reporting managers, HR, and senior leadership. The emphasis is on visual, at-a-glance summaries rather than raw data, so that decisions about coaching, budget, and resourcing can be made quickly.

US-4.1 As a manager, I want to view a heatmap of my team's skill gaps, so that I can prioritize coaching and resourcing decisions.

US-4.2 As an HR admin, I want to export competency and training-completion reports, so that leadership can review workforce-readiness metrics.

US-4.3 As a manager, I want to see each team member's learning-path completion status, so that I can follow up on overdue activities.

Epic 5 — Learning Path & Notifications

The final epic keeps employees engaged after a recommendation is made. It sequences approved training into a coherent role-based learning path and uses automated, configurable reminders to reduce the drop-off that typically occurs between assigning and completing training.

US-5.1 As an employee, I want to follow a role-based learning path with a clear sequence of courses, so that I develop competencies in the right order.

US-5.2 As an employee, I want to receive automated reminders for assigned but incomplete training, so that I don't miss deadlines.

US-5.3 As an HR admin, I want to configure notification frequency and channel (email/portal), so that reminders match organizational communication norms.

3 · Acceptance Criteria (Given–When–Then)

Acceptance criteria are defined for the user stories selected into Sprint 1 (Section 5).

US-1.1 — Employee submits certifications and project experience

Given the employee is logged in and on the “My Profile” page

When they upload a certification file and enter project details, then click Save

Then the profile is updated instantly, the file is stored, and a confirmation banner is shown.

US-2.1 — System compares profile against role competency matrix

Given an employee's profile is complete and a competency framework exists for their role

When the gap-analysis job runs (on-demand or nightly batch)

Then a skill-gap report is generated listing each competency as Met / Partial / Gap with a numeric deficiency score.

US-3.1 — System recommends ranked training for a gap

Given at least one competency is flagged as “Gap” for an employee

When the recommendation engine executes after gap analysis completes

Then the top 3 relevant courses are displayed, ranked by expected competency impact, with course duration and provider shown.

US-4.1 — Manager views team skill-gap heatmap

Given a manager opens the Team Dashboard for a direct-report group

When the dashboard loads

Then a color-coded heatmap (competency × employee) renders within 3 seconds, with red/amber/green severity indicators.

US-5.2 — Automated reminder for incomplete training

Given a training item assigned to an employee is still incomplete 3 days before its due date

When the daily notification scheduler runs

Then the employee receives an email and in-portal alert listing the pending item and days remaining.

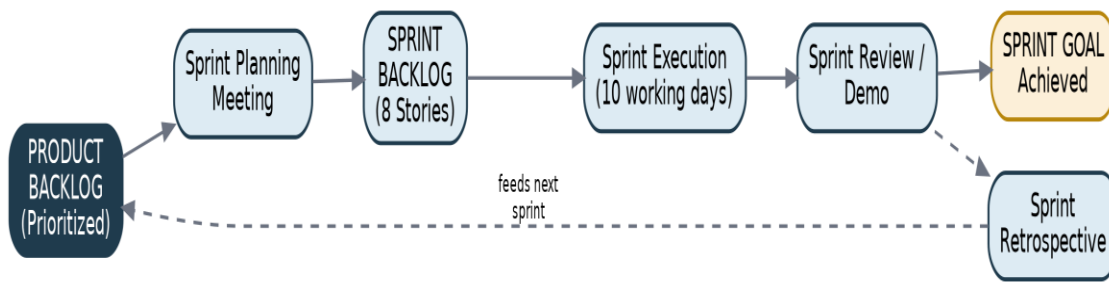
4 · Product Backlog Prioritization

Prioritized using a weighted score of Business Value (BV, 1–10), Customer/User Need (CN, 1–10) and inverse Dependency Risk (DR, 1–10, higher = fewer blockers), Priority Score = $(BV \times 0.4) + (CN \times 0.35) + (DR \times 0.25)$.

ID	Backlog Item	BV	CN	DR	Priority
US-1.1	Submit certifications & project experience	8	9	8	8.3 — High
US-1.2	Auto-import performance review scores	7	6	5	6.4 — Med
US-2.1	Compare profile vs competency matrix	10	9	7	9.1 — High
US-2.2	Define/edit competency frameworks	9	7	6	7.9 — High
US-3.1	Ranked training recommendations	9	9	6	8.4 — High
US-3.2	Accept / defer recommended course	6	8	9	7.4 — Med
US-4.1	Team skill-gap heatmap	8	7	7	7.4 — Med
US-4.2	Export competency reports	6	5	8	6.15 — Med
US-5.1	Role-based learning path sequence	7	8	5	7.05 — Med
US-5.2	Automated incomplete-training reminders	5	7	9	6.7 — Med

Dependency note: US-2.1 depends on a competency framework existing (US-2.2), and US-3.1 depends on gap results from US-2.1 — this dependency chain is why all three are pulled into Sprint 1 together despite US-2.2 scoring lower individually.

5 · Sprint Backlog — Sprint 1



Sprint 1 pulls the five highest-priority, dependency-connected stories from the top of the backlog and breaks each into actionable tasks.

Story	Task Breakdown	Owner Role
US-1.1	Design profile schema · Build upload API · Build “My Profile” UI · Unit tests	Frontend + Backend Dev
US-2.2	Model competency-framework schema · Build admin CRUD screen · Seed 3 role frameworks	Backend Dev + HR Analyst
US-2.1	Build comparison algorithm · Create gap-scoring logic · Build skill-gap report API	Backend Dev + Data Analyst
US-3.1	Design ranking heuristic · Map course catalogue to competencies · Build recommendation API	Backend Dev + ML Contributor
US-4.1	Design heatmap component · Build dashboard aggregation query · Integrate with UI	Frontend Dev + Backend Dev

6 · Story Point Estimation (Fibonacci / Planning Poker)

Story	Description	Points	Justification
US-1.1	Submit certifications & experience	5	Moderate complexity: file upload, validation, storage; no external dependency, team has done similar CRUD forms before.
US-2.2	Define/edit competency frameworks	5	New data model plus admin UI; scope is well understood, low ambiguity.
US-2.1	Compare profile vs competency matrix	8	Core algorithmic logic, needs careful scoring design and testing across multiple role types; medium-high uncertainty.
US-3.1	Ranked training recommendations	8	Depends on US-2.1 output; ranking heuristic and catalogue mapping add integration complexity.
US-4.1	Team skill-gap heatmap	5	Primarily a visualization + aggregation task; pattern is familiar to the team from prior dashboards.

Total committed effort: 31 story points. Planning Poker was used with the whole team voting simultaneously; where estimates diverged (US-2.1: votes of 5, 8, 13), the team discussed unknowns in the scoring algorithm and converged on 8 as the consensus estimate.

7 · Sprint Goal & Expected Deliverables

Sprint Goal: Enable an employee to build a competency profile, have it automatically compared against their role's requirements, and receive a ranked training recommendation — visible end-to-end on the manager's dashboard, so that the team can demonstrate a complete, working slice of the skill-gap analysis pipeline at the sprint review.

Expected Deliverables at Sprint End

- A working “My Profile” module where employees submit certifications and project experience.
- A configurable competency-framework module used by the gap-analysis engine.
- A functioning skill-gap comparison engine producing a Met / Partial / Gap report per employee.
- A ranked training recommendation list generated from identified gaps.
- A manager-facing skill-gap heatmap dashboard for the reporting team.
- Sprint Review demo covering the full flow: profile → gap analysis → recommendation → dashboard.
- Sprint Retrospective notes feeding refinement of Sprint 2 backlog (notifications & learning-path sequencing).

8 · Definition of Done

A user story is only marked complete when it satisfies every condition below. This checklist was agreed by the team during Sprint Planning and applies uniformly to all five stories committed in Sprint 1.

- Code is written, peer-reviewed, and merged into the main development branch without unresolved comments.
- All acceptance criteria defined in Section 3 pass in a manual or automated walkthrough.
- Unit tests are written for new logic and the existing test suite passes with no regressions.
- The feature is deployed to the staging environment and is demonstrable to the Product Owner.
- Relevant documentation (API contracts, schema changes, admin instructions) is updated.
- No known critical or high-severity defects remain open against the story.

9 · Risks & Assumptions

The team identified the following risks during backlog refinement, along with the assumptions the sprint plan currently relies on. These are revisited at each Daily Scrum and formally reviewed at the Sprint Retrospective.

Risk / Assumption	Potential Impact	Mitigation
Competency framework data is incomplete for some roles	Gap analysis produces inaccurate or partial results	HR to finalize frameworks for the 3 pilot roles before Sprint 1 starts
Performance-review import format varies across departments	Delays to US-1.2 integration	Build an adapter layer; fall back to manual CSV upload for Sprint 1
Training catalogue mapping is manual and time-consuming	Recommendation quality depends on mapping completeness	Scope Sprint 1 catalogue to a curated subset of 25 courses

Assumption: employees have valid HRMS accounts already	Login/identity issues could block adoption	Confirm SSO integration with IT before Sprint 1 demo
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Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25